Project Title
Near-Duplicate Question Finder for Q&A Platforms

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ABSTRACT

Q&A platforms often contain multiple questions that are identical or highly similar but expressed using different words or sentence structures. Manually identifying such near-duplicate questions is time-consuming and inefficient for large datasets. This project proposes a Near-Duplicate Question Finder using k-gram shingling, double polynomial hashing, Locality-Sensitive Hashing (LSH), and exact common-substring verification. Each question is divided into overlapping k-grams to capture local textual patterns. These k-grams are fingerprinted using two polynomial hash functions to reduce hash collisions. LSH then efficiently shortlists question pairs that are likely to be near-duplicates, avoiding expensive comparisons between every pair. The shortlisted candidates are finally verified using exact common-substring analysis to remove false positives and confirm genuine similarities. The system evaluates precision before and after verification, and the precision gain contributed by verification is reported. This approach aims to provide an accurate and scalable solution for duplicate-question detection on Q&A platforms and online knowledge repositories.

Signature of the Guide